Final Year Project Proposal

Cotton Leaf Disease Detection from Leaf Photos for Punjab Farmers Using Deep Learning

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Problem statement

Farmers growing cotton in Punjab often cannot get a timely, expert diagnosis of leaf diseases, so treatment starts late and yields suffer. Agriculture extension officers are few relative to the number of farms, and many farmers rely on guesswork or advice from local pesticide shops.

Deep learning models can recognise disease symptoms from a leaf photo, but most published work is trained on clean laboratory images and on crops that are not grown here. This project builds and evaluates a leaf-disease classifier for cotton, the main cash crop of southern Punjab, and wraps it in a simple app that anyone with a phone camera can use.

Objectives

1. Build a labelled image dataset of cotton leaves covering 3 classes (bacterial blight, leaf curl virus, healthy), combining public data with photos taken in local fields.
2. Train and compare two transfer-learning models (MobileNetV2 and EfficientNet-B0) using accuracy, precision, recall and F1-score.
3. Test the best model on a held-out set of real field photos and report how much accuracy drops compared with clean public images.
4. Develop a web app where a user uploads a leaf photo and receives the disease name, a confidence score and short treatment advice in English or Urdu.
5. Document the system, results and limitations in a final report and demonstrate it live.

Scope

The system covers one crop, cotton, with 3 classes, delivered as one web app.

Crop	Classes
Cotton	Bacterial blight, leaf curl virus, healthy

The final class list depends on which diseases have enough usable images and will be confirmed with the supervisor. Out of scope: other crops (including wheat), insect pest detection, a native mobile app, voice output and weather alerts. These are listed as future work.

Methodology

The project follows five stages, with a working end-to-end version built early and improved afterwards.

1. **Data collection:** Gather public cotton leaf datasets (for example from Kaggle) and check their licences and label quality. Add photos taken in local fields with a phone camera, under varied light and backgrounds. Ask an agriculture student or extension officer to verify a sample of labels.
2. **Preprocessing:** Remove duplicates and mislabelled images, resize to 224x224, and split into train, validation and test sets. Keep the field photos as a separate test set so they are never seen during training. Apply augmentation (rotation, flip, brightness, zoom) to the training set only.
3. **Model training:** Fine-tune MobileNetV2 and EfficientNet-B0 pretrained on ImageNet, using Google Colab or Kaggle GPUs. Tune learning rate, batch size and number of unfrozen layers.
4. **Evaluation:** Compare both models on accuracy, precision, recall, F1-score and confusion matrices. Report results separately for public images and field photos.
5. **Deployment:** Wrap the best model in a Streamlit web app that returns the disease, a confidence score and stored treatment advice. Low-confidence predictions show a warning asking the user to retake the photo or consult an expert.

Tools and technologies

Area	Tools
Language	Python 3
Deep learning	PyTorch and torchvision, with pretrained ImageNet weights
Training environment	Google Colab and Kaggle notebooks (free GPU)
Data handling	NumPy, pandas, Pillow
Evaluation	scikit-learn, Matplotlib
Application	Streamlit web app
Testing	pytest, Playwright
Version control	Git and GitHub

Expected outcomes and deliverables

- A cleaned, documented dataset of cotton leaf images, including locally collected photos
- Two trained models with a side-by-side performance comparison
- A web app that returns disease name, confidence and treatment advice
- A field-test report on cotton leaves showing accuracy on real photos and known weak cases
- Source code on GitHub, SRS and SDD documents, a final report and a live demo

Timeline

The work is split across two semesters, with a working prototype by the end of Semester 7. Weeks are counted from the start of each semester.

Semester	Weeks	Milestone
7	1-3	Finalise topic and classes, write proposal, review 8-10 papers
7	4-8	Collect and clean data, set up the training pipeline in Colab
7	9-14	Train and compare the two models, first end-to-end prototype
8	1-6	Improve the model and build the web app
8	7-11	Field testing with real photos, fix weak classes

Semester	Weeks	Milestone
8	12-16	Final report, presentation and demo practice

Risks and mitigation

Risk	Mitigation
Too few public images for some diseases	Drop that class or collect more local photos; confirm the final class list by week 8
Model works on public images but fails on field photos	Keep a separate field test set, use augmentation, add local photos to training
Wrong labels or treatment advice	Have an agriculture student or extension officer check labels and advice
Falling behind as a solo student	Build a rough end-to-end version early, keep a weekly progress log, cut optional features first
Limited laptop power	Train on Colab or Kaggle GPUs and keep models small

Student signature	Supervisor signature (Dr Munawar)	Date